

```
In[32]:= ClearAll["Global`*"];
```

```
podaci = {  
  {1, "n/a"},  
  {x^n, "n != -1"},  
  {1/x, "x != 0"},  
  {E^x, "n/a"},  
  {a^x, "a > 0, a != 1"},  
  {Sin[x], "n/a"},  
  {Cos[x], "n/a"},  
  {Tan[x], "x !=  $\pi/2 + k\pi$ "},  
  {Cot[x], "x !=  $k\pi$ "},  
  {1/Cos[x]^2, "x !=  $\pi/2 + k\pi$ "},  
  {1/Sin[x]^2, "x !=  $k\pi$ "},  
  {1/Sqrt[1 - x^2], "|x| < 1"},  
  {1/(1 + x^2), "n/a"},  
  {1/Sqrt[a^2 - x^2], "|x| < |a|"},  
  {1/(a^2 + x^2), "a != 0"},  
  {1/(x^2 - a^2), "|x| != |a|"},  
  {Sinh[x], "n/a"},  
  {Cosh[x], "n/a"},  
  {Log[x], "x > 0"},  
  {ArcSin[x], "|x| <= 1"},  
  {ArcTan[x], "n/a"},  
  {x * E^x, "n/a"}  
};
```

```
obradiRed[{f_, uslov_}] := Module[{intExpr, resExpr},  
  intExpr = TraditionalForm[HoldForm[Integrate[f, x]]];  
  resExpr = TraditionalForm[Simplify[Integrate[f, x]] + C];  
  {TraditionalForm[f], intExpr, resExpr, Style[uslov, Italic, 11, GrayLevel[0.3]]}  
];
```

```
tabelaRedovi = Map[obradiRed, podaci];
```

```
zaglavlje = {
```

```

Style["Funkcija f(x)", Bold, 13, Darker[Blue]],
Style["Integral  $\int f(x) dx$ ", Bold, 13, Darker[Blue]],
Style["Rezultat (Neodređeni integral)", Bold, 13, Darker[Blue]],
Style["Opombe / Pogoji", Bold, 13, Darker[Blue]]
};

```

```

tabelaIntegrala = Grid[
  Prepend[tabelaRedovi, zaglavlje],
  Frame → All,
  FrameStyle → Directive[Thickness[1], GrayLevel[0.75]],
  Background → {
    None,
    {RGBColor[0.9, 0.93, 0.98],
     Table[If[EvenQ[i], White, RGBColor[0.96, 0.97, 1.0]], {i, Length[podaci]}]}
  },
  ItemMargins → {{1.5, 1.5}, {0.8, 0.8}},
  Alignment → {Center, Center}
];

```

```

dokument = Column[{
  Style["Tabela nedoločenih integralov", Bold, 22, Darker[Blue]],
  Spacer[5],
  Style["Projektna naloga pri predmetu Računalniška orodja v matematiki (FMF)",
    12, Italic, Darker[Gray]],
  Spacer[15],
  tabelaIntegrala,
  Spacer[15],
  Style["Generirano s pomočjo programa Wolfram Mathematica", 10, Italic, Gray]
}, Alignment → Center];

```

```

Export["Tabela_Nedolocnih_Integralov.pdf", dokument];
dokument

```

Out[40]=

Tabela nedoločenih integralov

Projektna naloga pri predmetu Računalniška orodja v matematiki (FMF)

Funkcija $f(x)$	Integral $\int f(x) dx$	Rezultat (Neodređeni integral)	Opombe / Pogoji
1	$\int 1 dx$	$C + x$	n/a
x^n	$\int x^n dx$	$C + \frac{x^{n+1}}{n+1}$	$n \neq -1$
$\frac{1}{x}$	$\int \frac{1}{x} dx$	$C + \log(x)$	$x \neq 0$
e^x	$\int e^x dx$	$C + e^x$	n/a
a^x	$\int a^x dx$	$\frac{a^x}{\log(a)} + C$	$a > 0, a \neq 1$
$\sin(x)$	$\int \sin(x) dx$	$C - \cos(x)$	n/a
$\cos(x)$	$\int \cos(x) dx$	$C + \sin(x)$	n/a
$\tan(x)$	$\int \tan(x) dx$	$C - \log(\cos(x))$	$x \neq \pi/2 + k\pi$
$\cot(x)$	$\int \cot(x) dx$	$C + \log(\sin(x))$	$x \neq k\pi$
$\sec^2(x)$	$\int \sec^2(x) dx$	$C + \tan(x)$	$x \neq \pi/2 + k\pi$
$\csc^2(x)$	$\int \csc^2(x) dx$	$C - \cot(x)$	$x \neq k\pi$
$\frac{1}{\sqrt{1-x^2}}$	$\int \frac{1}{\sqrt{1-x^2}} dx$	$C + \sin^{-1}(x)$	$ x < 1$
$\frac{1}{x^2+1}$	$\int \frac{1}{1+x^2} dx$	$C + \tan^{-1}(x)$	n/a
$\frac{1}{\sqrt{a^2-x^2}}$	$\int \frac{1}{\sqrt{a^2-x^2}} dx$	$\tan^{-1}\left(\frac{x}{\sqrt{a^2-x^2}}\right) + C$	$ x < a $
$\frac{1}{a^2+x^2}$	$\int \frac{1}{a^2+x^2} dx$	$\frac{\tan^{-1}\left(\frac{x}{a}\right)}{a} + C$	$a \neq 0$
$\frac{1}{x^2-a^2}$	$\int \frac{1}{-a^2+x^2} dx$	$C - \frac{\tanh^{-1}\left(\frac{x}{a}\right)}{a}$	$ x \neq a $
$\sinh(x)$	$\int \sinh(x) dx$	$C + \cosh(x)$	n/a
$\cosh(x)$	$\int \cosh(x) dx$	$C + \sinh(x)$	n/a
$\log(x)$	$\int \log(x) dx$	$C + x(\log(x) - 1)$	$x > 0$
$\sin^{-1}(x)$	$\int \sin^{-1}(x) dx$	$C + \sqrt{1-x^2} + x \sin^{-1}(x)$	$ x \leq 1$
$\tan^{-1}(x)$	$\int \tan^{-1}(x) dx$	$C - \frac{1}{2} \log(x^2+1) + x \tan^{-1}(x)$	n/a
$e^x x$	$\int e^x x dx$	$C + e^x (x - 1)$	n/a

Generirano s pomočjo programa Wolfram Mathematica